

Homework 1 – Evaluating Expressions

- Complete the table of inputs and outputs for the following code:

```
a = int(input('Enter a number: '))
b = a
while b >= 0:
    if b % 2 == 0:
        b //= 2
    else:
        b -= 2*(a+2)
    if b == 0:
        a = b = 2
        break
print(a*-b)
```

input	output
0	
1	
2	
4	
5	
10	
15	
-9	

Homework 2 – Leap Years

- Most years that are divisible by 4 are leap years. Years that are divisible by 100 are only leap years if they are divisible by 400.
 - For example, the years 1700, 1800, and 1900 are not leap years, but the years 1600 and 2000 are.
- **Write a program that prints out the leap years from 8 AD (the first AD leap year) up until this year.**
- Use loops, `if/else`, `range()`, `continue`, `break`, or `pass`
 - Your solution may use some or all of these elements

Homework 3 – Amount Due, Pt. 3

- **Extend** the program you wrote last week for the problem Amount Due to give the user the ability to do all of the following:
 - **add more items to the purchase**
- When no more items are added by the user, **output the final total on screen.**
 - As a challenge, add a per item breakdown and/or
 - retain the ability to use a card for a 3% fee